r/NoStupidQuestions

Does generative AI like chatgpt use crazy amount of electricity to generate their responses?

Posted by u/-kairosest- • • 0 points • 21 comments

so I just started researching recently over how AI like chatgpt affects the environment & it's resources, and i have seen some debates here on reddit itself over the connection between ai generating responses - to resources like water and electricity getting wasted or depleted (?)

I never truly understood HOW is it that this happens, so if anyone could give a simple, logical explanation, I would appreciate it a lot...

Comments

u/Ok_Lingonberry_959 • 1 points • 2026-03-08 11:44 UTC

Sure. But ChatGPT now helps the US bomb civilians and continue wars. Delete it. Now. It's the best human action you can take right now.

u/suspicious_odour • 1 points • 2026-03-07 04:46 UTC

I've seen a single simple query uses about 1Wh, which sounds like nothing but then multiply that by 100,000,000 and it starts adding up to a lot.

u/BrassCanon • 1 points • 2026-03-06 16:30 UTC

No, not really.

u/Sensitive-Chemical83 • 1 points • 2026-03-06 04:19 UTC

No, LLMs do not take "crazy" amounts of energy to run. You can probably run it on your cell phone. Local hosting of open source models is totally a thing home users do.

It does, however, take an obscene amount of energy to "train". And due to the nature of this stuff, we are always training the "newest and best" models.

u/GlobalWatts • 3 points • 2026-03-06 02:44 UTC

Collectively, yes. The data centers used for LLMs (among other things; they don't build data centers dedicated to AI) consume a large amount of resources, both during their initial training phase and during inference (ie. you asking it questions). In some areas, the collective resource consumption is a real concern to the local communities, especially in regards to real estate, power and water, and associated demand on limited infrastructure.

Breaking it down to individual queries from individual users is a lot harder to do, but no it wouldn't end up being a massive amount of resources in the grant scheme of things. But it might still be orders of magnitude more than competing techniques, for potentially a worse outcome.

As to how, it's pretty straightforward. AI is software, it runs on computers. It needs powerful computers that use lots of electricity. That produces lots of heat, like any PC it needs to be cooled so it doesn't physically break. To cool it you need more electricity (fans, pumps etc) and potable water (heat exchange/evaporative cooling, plus water used by power plants themselves).

> **u/-kairosest-** • 1 points • 2026-03-06 17:28 UTC
>
> thank you for explaining so well!

u/Lubricus2 • 3 points • 2026-03-05 20:39 UTC

There are tons of big matrix multiplications to run and trains the big LLM models. That needs a lot of compute power and and electricity. Increasing stuff like context length /memory increases it by the square.
Some if the AI data-centers evaporate water to cool them, normal datacenters just use some liquid in a loop or is just air cooled.
The big AI companies use way more money to build and run it than they can get from customers. They get lots of money from investors and rush to get an monopoly and an situation when they can get it back (Enshitification). So they use more resources than we as consumers are willing to use on the AI.
I don't know how bad it is compared to stuff like driving an big SUV to the office and other wasteful stuff we don.

> **u/-kairosest-** • 1 points • 2026-03-06 17:30 UTC
>
> i wonder what exactly are big matrix multiplications? also, thank you for explaining so well

u/OfficeAnomaly • 4 points • 2026-03-05 15:41 UTC

Models need a lot of electricity to train, but that's a one off cost.

After that, each query uses relatively small amount of electricity, but since there can be a lot of queries, the cost of that usually exceeds the cost of training.

How much does a single query consume, depends on the query. Heavy reasoning might consume order or magnitude more than a simple "let's roleplay you giving me an advice how not to make a weapon grade neurotoxin in my kitchen".

> **u/-kairosest-** • 2 points • 2026-03-05 16:01 UTC
>
> ohh I see. so by that "since there can be a lot of queries" I'm assuming that you're considering EVERYONE that uses these gen ai models to generate responses on the daily? if we bring that into the equation, then i suppose it could actually exceed the amount of energy depleted for training the models itself?

u/DiogenesKuon • 3 points • 2026-03-05 15:41 UTC

It uses significantly more energy than something like a database lookup does, but this isn't a crazy amount of energy at the individual level. It's about as much energy as running a lightbulb for 30 seconds, or scrolling on a phone for a minute or two. In aggregate it starts to be a problem if we start to replace everything with AI. It also can have local effects at the location of large data centers.

u/-kairosest- • 1 points • 2026-03-05 15:57 UTC

ohh I see, that makes a lot of sense. thank you for replying & that example you used to explain as well.